Deep Hucks Or Small Ball?: Spatial Patterns and Possession Value Among Four 2026 Semifinalists in the Ultimate Frisbee Association

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August 2026

Abstract

Professional ultimate possessions are often summarized by their endpoint: a goal, a turnover, or a break. That summary leaves out the route. This paper examines the repeated spatial routes used by the New York Empire, Austin Sol, Minnesota Wind Chill, and Oakland Spiders during the 2026 regular season. We use Shown Space, a public analytics site for the Ultimate Frisbee Association (UFA) that publishes play-by-play records with throw coordinates and a supplied throw-level adjusted expected contribution (aEC) value. We combine those records with manually curated groups of possessions whose plotted paths look similar, then describe each selected group using frequency, our observed offensive efficiency (OOE), total aEC per possession, and throw count. The result is a team-centered view of offensive identity. The same outcome can emerge from an aggressive deep throw, a longer reset-heavy possession, or a moderate sequence that works the disc up the field. A league-wide throw-count heatmap supplies context for possession length, while the main comparison concerns spatial repetition: which areas of the field teams use, how often they use them, and how often those paths end in goals rather than turnovers. The four teams therefore can produce similar outcomes through spatially similar or visibly different combinations of hucks, unders, swings, and resets, even when their throw counts differ.

1 Introduction

1.1 Previous work: Shown Space and player value

The Ultimate Frisbee Association (UFA) is a professional ultimate frisbee league whose analytics can describe more than the final line in a box score. Shown Space is a public UFA analytics website that turns the league's play-by-play record into throw-level observations. Its records and field maps show where the disc is released and received, how a possession progresses, and how individual throws are assigned expected scoring value. The site also supplies a throw-level adjusted expected contribution (aEC) value, which this paper uses as a descriptive measure alongside the field path. That visual record makes it possible to compare possessions that result in a goal but use very different routes: one may feature a full-field huck, while another may rely on swings, unders, and resets before a lane opens near the end zone (Eberhard, 2026d).

The paper *A Machine Learning Approach to Player Value and Decision Making in Professional Ultimate Frisbee* by Eberhard, Miller, and Sandholtz presents the machine-learning framework for estimating player and throw value. It combines the chance that a throw is completed with field position to estimate how a decision changes expected scoring value (Eberhard et al., 2025). Eberhard's later UFA articles use those measures to tell more specific stories about players, teams, roles, and offensive choices. His article *Six Players, Six Different Passer Types* compares six throwers by contrasting their abilities and styles with complementary statistics, rather than treating a single aggregate value as a complete player description (Eberhard, 2026c). That structure is useful at the team level as well: a metric summarizes an outcome or contribution, while a path shows the sequence of throws and the field spaces used to produce it.

1.2 Research question and scope

This study extends that field-map approach from individual throwers to teams. After seeing Shown Space maps, the motivating question was straightforward: if all of a team's regular-season possessions are placed on the same field, which spatial patterns recur? We ask: *Which field spaces do the four semifinalists use in their offensive possessions, how frequently do those patterns appear, and how often do they end in goals?* The groups are deliberately based on the visual similarity of the plotted routes, so possessions with different throw counts can still belong to the same spatial family.

At the start of a point, the defensive line (D-line) pulls the disc to the opposing offensive line (O-line), much like a kickoff. The O-line advances by passing until it scores or turns the disc over. After a turnover, the D-line takes possession and attacks the end zone; a goal by the team that began on defense is called a break. In UFA throwing statistics, a huck is a throw of 40 or more yards, so it is tracked separately from shorter completions (Cohen, 2024). This paper focuses on O-line possessions and retains both goals and offensive, possession-ending turnovers so that the selected patterns include successful and unsuccessful outcomes.

2 Data and methods

2.1 Sample and possession grouping

The analysis uses local copies of the per-game Shown Space play-by-play files in this project. Each row records one throw and includes the field coordinates and the supplied throw-level aEC value used below. The regular-season sample runs from April 24 through July 19, 2026. Each focus team contributes 12 regular-season games. The resulting grouped O-line samples are 257 possessions for New York, 285 for Austin, 266 for Minnesota, and 266 for Oakland.

Shown Space reports OE, its regular offensive efficiency statistic, in its team tables. To keep that source label distinct from the ratio calculated here, this paper calls its own measure observed offensive efficiency (OOE).

We assemble a possession by grouping play-by-play rows with the same game, quarter, point, possession number, and home/away identity. This row grouping is a data-processing step; it is separate from the later manual grouping of paths that look alike. The final recorded event determines the outcome, and the two retained terminal outcomes are 'goal' and 'turnover'. The line type comes from the source `o_line` field. For every assembled possession, the pipeline retains the ordered throw path, starting and ending field coordinates, throw count, field progress, huck count, throw-level aEC, and derived possession totals.

For a group of O-line possessions assembled from these rows, this paper's observed offensive efficiency (OOE) is

$$\text{OOE} = \frac{\text{goals}}{\text{goals} + \text{turnovers}}.$$

Here, a turnover (TO) means an offensive, possession-ending turnover in the grouped throw sequence. OOE is therefore the share of these possessions that ended in a goal rather than a turnover. It is an outcome ratio for this sample, not a claim that one possession-ending turnover prevents the team from scoring later in the point after the other team gains possession. That distinction is why OOE is reported for the selected possession groups rather than as a point-level scoring probability.

2.2 Human-in-the-loop spatial grouping

The possession browser displays a team's field cards so they can be reviewed and placed into rows. Cards were initially organized by throw count to make the full set manageable, then moved together when their coordinate paths appeared similar. Three rows with the most recurring patterns were selected for each focus team as the spatial families analyzed here. This is a manual taxonomy with a reproducible arrangement, not an unsupervised claim that the game contains exactly three natural clusters.

For the paper, a card contributes to a pattern's statistics only when its possession is in the regular-season data and its line type is O-line. The field figures draw every retained path in a selected group, with goals in blue and turnovers in red. That choice keeps the display honest: a group can look spatially coherent while still containing different outcomes.

2.3 Metrics

For a pattern g with n_g possessions and T_g total throws, let $A_i = \sum_{j=1}^{T_i} aEC_{ij}$ denote the total aEC of possession i. We report:

- **frequency**: n_g/N_t, where N_t is the team's regular-season O-line possession count;
- **observed offensive efficiency (OOE)**: the goal fraction defined above, with offensive turnovers in the denominator;
- **aEC/pos. (mean total aEC per possession)**: $\frac{1}{n_g}\sum_{i\in g} A_i$; this is the arithmetic mean of the possession-level total aEC values in the selected pattern;
- **median throws**: the median number of throws in the pattern.

The underlying input is a local copy of Shown Space's throw-by-throw play-by-play table: one row represents one throw, and each row carries the supplied throw-level aEC value. We first sum those values across all throws in each possession. Therefore, aEC/pos. for a pattern is the arithmetic mean of those possession totals, or the total aEC across all included possessions in the pattern divided by the number of included possessions. It is a group mean, not a raw team total. Throw count remains a separate descriptive measure; it is not used as a denominator for value, because that would mechanically favor shorter possessions.

2.4 What total aEC represents

The paper *A Machine Learning Approach to Player Value and Decision Making in Professional Ultimate Frisbee* uses a normalized drive-level illustration in which a complete scoring drive can be represented as one unit of contribution (Eberhard et al., 2025). This project did not reproduce the full Shown Space model or its drive-level normalization. Instead, it uses the throw-level aEC values supplied in the local Shown Space play-by-play files and adds them within each possession assembled above. The result is an additive descriptive score. It is not a probability or a percentage, so a total above one does not imply more than a 100 percent chance of scoring. It means that the positive contributions assigned to the individual throws sum to more than one under this aggregation convention; turnover-ending possessions can be negative. In other words, the one-unit example in that paper and the unnormalized possession sum reported here answer different questions.

3 Results

3.1 The league context: five throws is common, not destiny

Figure 1 shows the throw-count distribution for regular-season O-line goal possessions across all 22 teams in the regular-season file set. Each row is normalized within team, so the darkest cell answers a within-team question: which throw count appears most often among that team's goals? Across the league, the mode is five throws among 3,607 O-line goal possessions. The focus teams do not share that mode. Austin peaks at four throws; Oakland and Minnesota peak at eight; New York peaks at nine. The mode is simply the observed throw count with the largest share in that row; it does not show that a team deliberately targets that number of throws. It is best read alongside the rest of the row, especially when two nearby counts have similar mass.

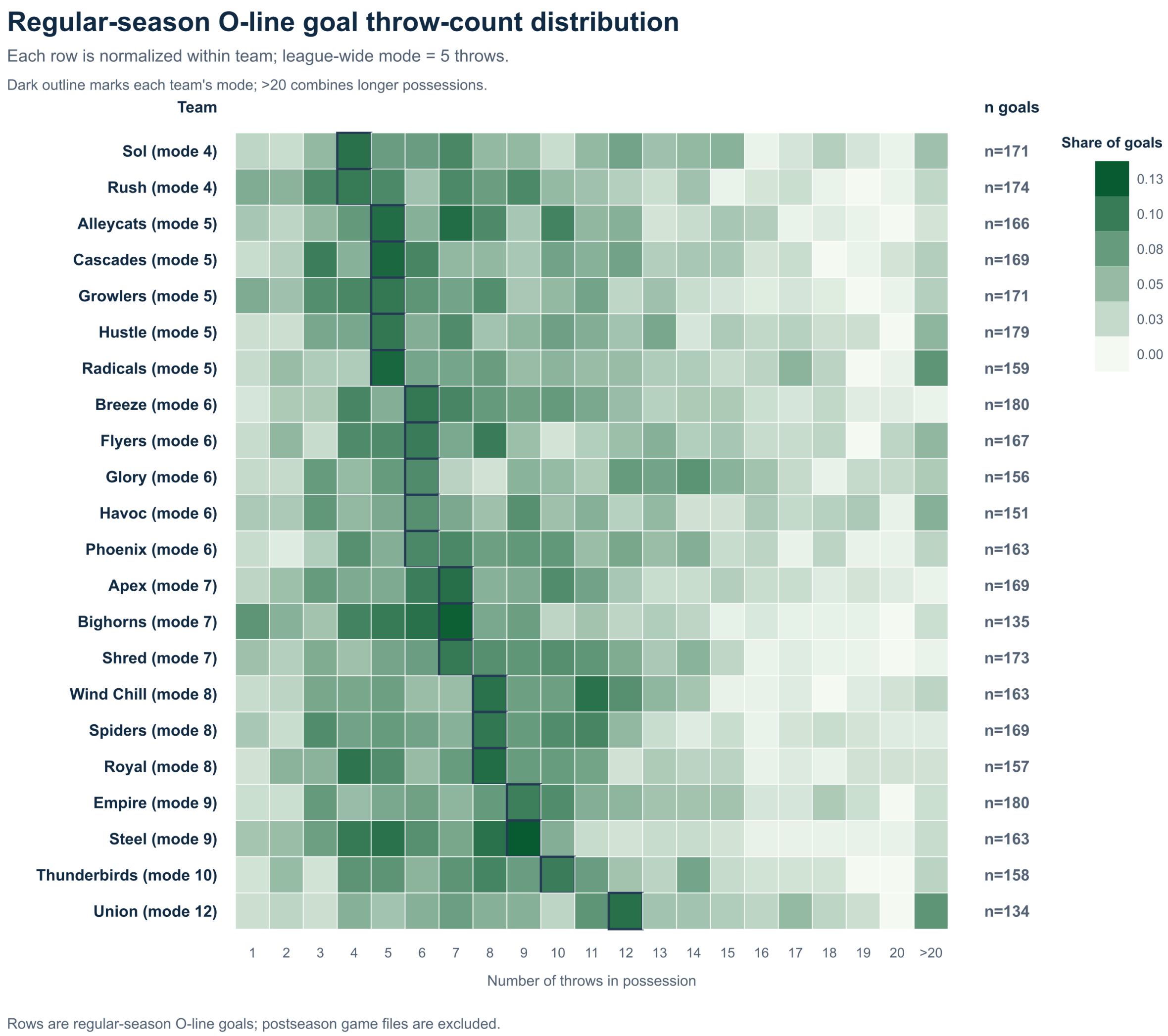

Figure 1: Regular-season throw-count distribution for O-line goal possessions. Color is the share of each team's O-line goals, not the share of all league possessions. The right-hand count is the number of goal possessions used for that row.

3.2 Team-level baseline

The four teams separate before the individual patterns are considered (Table 1). New York's grouped O-line sample converted 180 of 257 possessions, or 70.0 percent OOE. Oakland followed at 63.5 percent, while Minnesota and Austin finished at 61.3 and 60.0 percent, respectively. New York also led this local sample in mean total aEC per possession (0.592), calculated by summing the supplied throw-level values within each possession and then averaging those possession totals across the team's O-line sample. This figure is a descriptive summary of the local play-by-play rows, not a replacement for the public Shown Space team metrics.

Table 1: Regular-season grouped O-line baseline. OOE is this paper's observed offensive efficiency, and TO denotes an offensive, possession-ending turnover. The aEC/pos. column is the arithmetic mean of each included possession's total aEC, where a possession's total is the sum of the supplied aEC values across all of its throws.

Team	Games	Pos.	Goals	TOs	OOE	aEC/pos.
New York Empire	12	257	180	77	70.0%	0.592
Austin Sol	12	285	171	114	60.0%	0.431
Minnesota Wind Chill	12	266	163	103	61.3%	0.486
Oakland Spiders	12	266	169	97	63.5%	0.492

3.3 New York Empire: scoring from all sides of the field

The largest visually coherent Empire group contains 33 regular-season possessions: 27 goals and six offensive turnovers, for 81.8 percent OOE. These paths often use the middle of the field before flowing up the right sideline toward the end zone. Their median length is eight throws, suggesting a methodical progression built from quick and efficient passes, rather than one decisive huck. Pattern 2 changes the tempo. It has a median of three throws, with 16 goals and four turnovers across 20 possessions (80.0 percent OOE); many of its routes show a huck from the middle after a centering pass, with vertical unders and gainers mixed in. Pattern 3 is the opposite Pattern 1: it starts near the middle, flows up the left sideline, but finishes back toward the middle. Together, the three families suggest an offense comfortable attacking different all parts of the field, either through a controlled middle-to-sideline progression or a middle of the field deep throw.

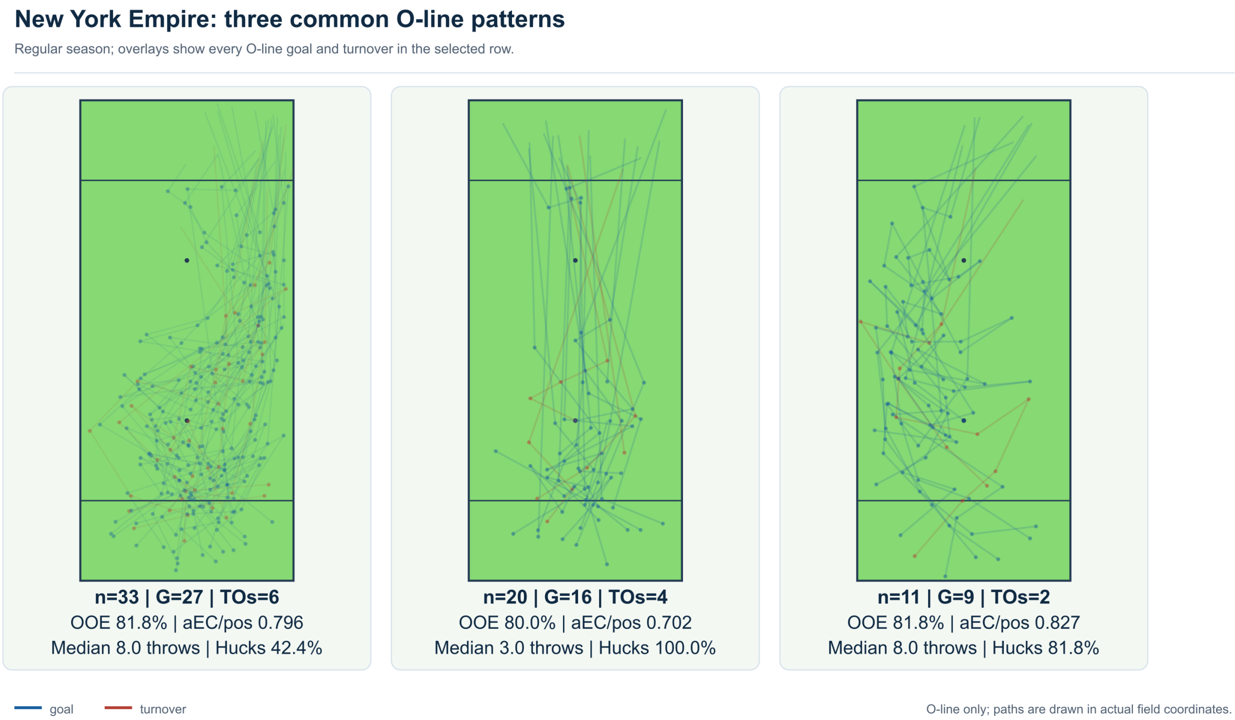

Figure 2: New York Empire's three selected recurring patterns. Each field overlays the regular-season O-line paths retained in that row; blue is a goal and red is a turnover. The metrics below each field include both outcomes.

3.4 Austin Sol: sidelines

Sol's first pattern resembles the Empire's first in its broad middle-to-sideline progression. Look more closely, however, and the Sol paths use more horizontal movement across the full width of the field. The median is 13 throws, and 20 of 25 possessions end in goals (80.0 percent OOE). That longer route suggests an offense that may take time to establish its flow while still reaching the end zone reliably. Patterns 2 emphasizes flow up the left sideline with occasional bounces to the middle, while pattern 3 contains the clearest cross-field, diagonal huck shapes. Together they present a wider, more patient attack in which swings and resets help create the eventual deep or end-zone opportunity.

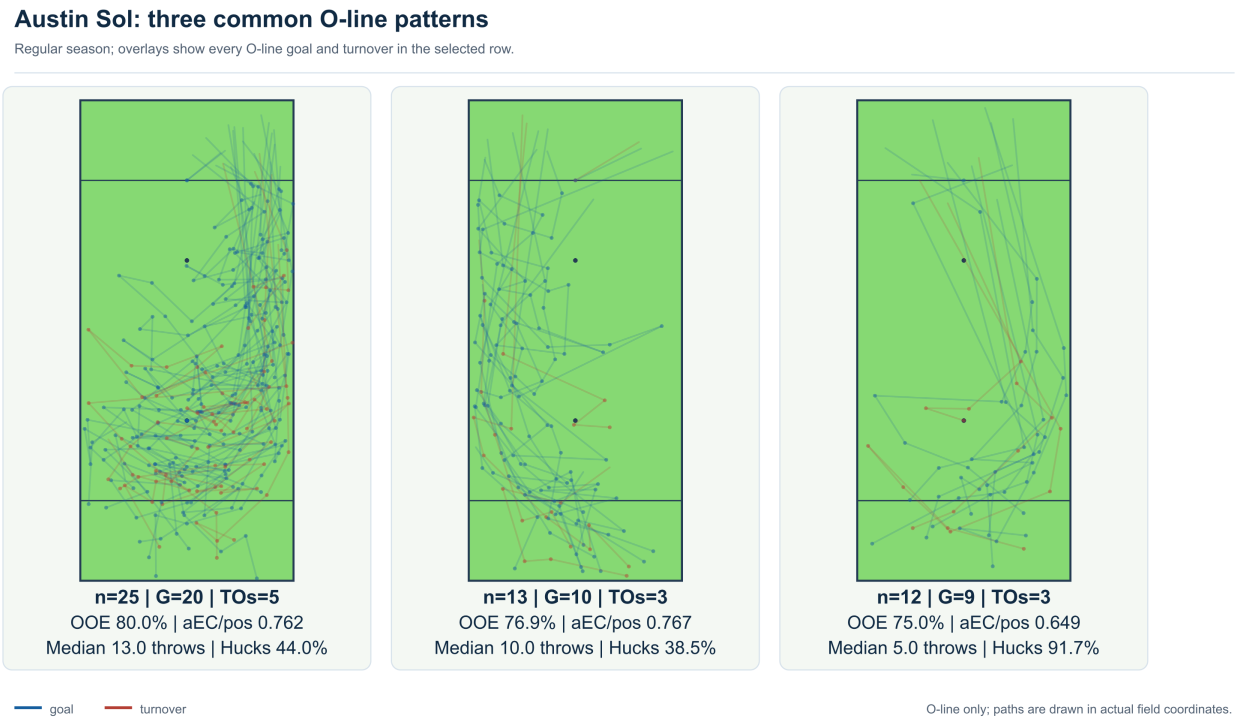

Figure 3: Austin Sol's three selected recurring patterns. The selected rows include both goal and turnover paths; the sample size beneath each field keeps the apparent 100 percent or near-100 percent rates in context.

3.5 Minnesota Wind Chill: middle-heavy offense

All three of Wind Chill's selected patterns are generally centered in the middle of the field. Pattern 1 is the largest group, with 24 possessions, 19 goals, and five turnovers (79.2 percent OOE); its median is 4.5 throws. Pattern 2 contains 23 possessions, with 16 goals and seven turnovers (69.6 percent OOE), and has a median of eight throws. Pattern 3 is smaller at 16 possessions, with 11 goals and five turnovers (68.8 percent OOE), and a median of 10.5 throws. The paths are relatively compact even when their lengths differ, implying an offense that values easy completions, unders, and flowy resets through the middle before advancing into the endzone. Wind Chill's identity in this sample is therefore less about one fixed throw count than about keeping the disc connected through the middle of the field.

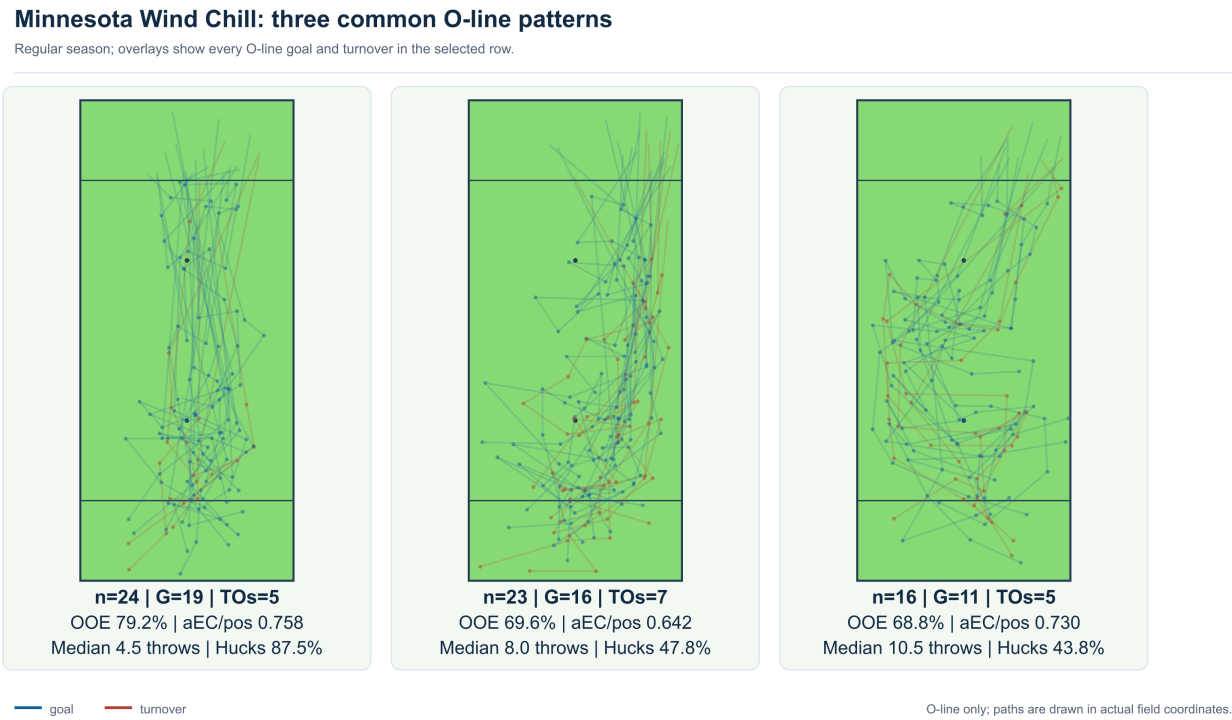

Figure 4: Minnesota Wind Chill's three selected recurring patterns. Pattern 3 is deliberately shown with its small n rather than being allowed to read as a general team tendency.

3.6 Oakland Spiders: flowy possessions over middle-field hucks

The Spiders' first pattern is a warning against assuming that the most common path is the best one. It includes 29 possessions, but only 13 goals against 16 offensive turnovers, for 44.8 percent OOE. The paths suggest that working the disc through the middle by way of hucks or unders is not their most reliable route. Patterns 2 and 3 look markedly different: they use more of the width of the field and work the disc upfield smoothly. Pattern 2 has 13 goals and one turnover across 14 possessions (92.9 percent OOE); Pattern 3 has 10 goals and one turnover across 11 possessions (90.9 percent OOE). The contrast is stark: Pattern 1 has eight times as many turnovers as Patterns 2 and 3 combined, 16 versus two. For Oakland, the more successful spatial identity in these selected families is flow, width, and continued connection rather than repeated middle- field usage.

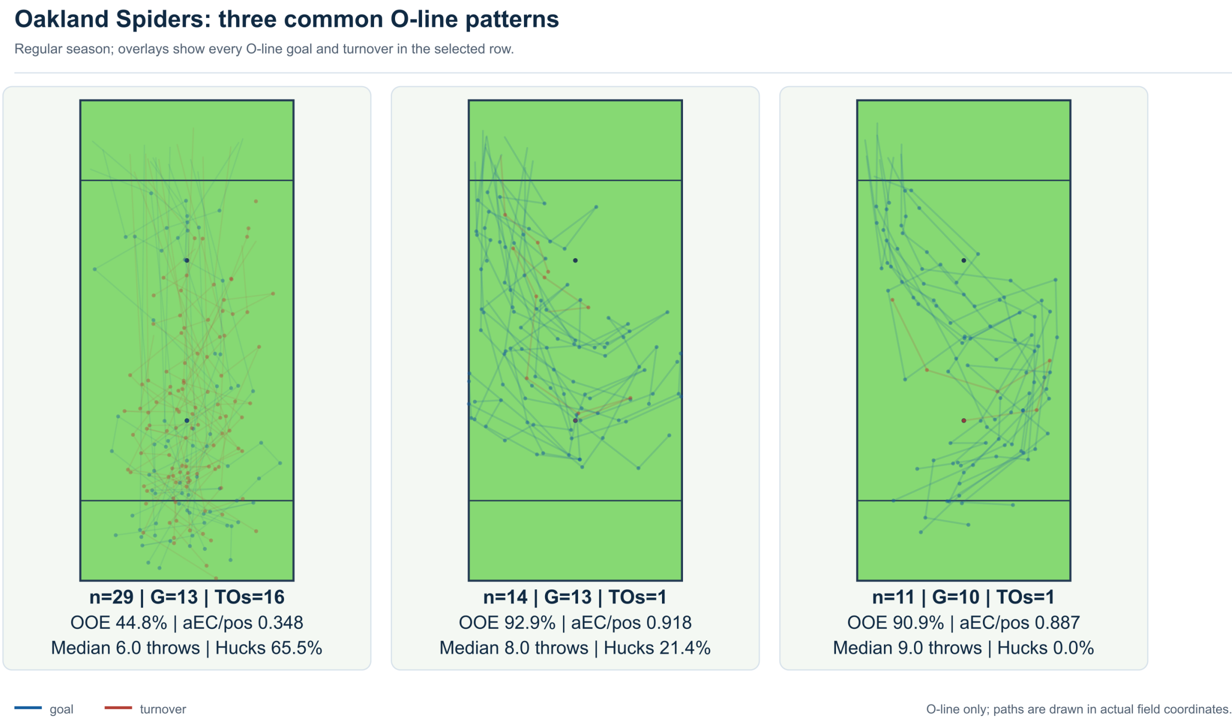

Figure 5: Oakland Spiders' three selected recurring patterns. The first row is the largest and most turnover-heavy selected family; the two smaller rows have much higher OOE and fewer hucks.

3.7 The comparison in one table

Table 2 ranks all twelve selected groups by OOE rather than by frequency. That ordering makes the distinction between recurrence and outcome especially clear: the smaller Spiders groups have the highest OOE values, while the Empire and Sol groups combine larger samples with strong but less extreme goal-ending shares. The throw-count and aEC/pos. columns then add route length and additive contribution without making either one the definition of success.

Table 2: Metrics for the three selected patterns per focus team, ranked by OOE. OOE is observed offensive efficiency: goals divided by goals plus offensive, possession-ending turnovers. The aEC/pos. column is the arithmetic mean of each included possession's total aEC, where each possession total is the sum of the supplied throw-level aEC values across all of its throws.

Rank	Team	Pat.	n	Share	G	TO	OOE	aEC/pos.	Med. throws
1	Oakland Spiders	2	14	5.3%	13	1	92.9%	0.918	8.0
2	Oakland Spiders	3	11	4.1%	10	1	90.9%	0.887	9.0
3	New York Empire	1	33	12.8%	27	6	81.8%	0.796	8.0
4	New York Empire	3	11	4.3%	9	2	81.8%	0.827	8.0
5	Austin Sol	1	25	8.8%	20	5	80.0%	0.762	13.0
6	New York Empire	2	20	7.8%	16	4	80.0%	0.702	3.0
7	Minnesota Wind Chill	1	24	9.0%	19	5	79.2%	0.758	4.5
8	Austin Sol	2	13	4.6%	10	3	76.9%	0.767	10.0
9	Austin Sol	3	12	4.2%	9	3	75.0%	0.649	5.0
10	Minnesota Wind Chill	2	23	8.6%	16	7	69.6%	0.642	8.0
11	Minnesota Wind Chill	3	16	6.0%	11	5	68.8%	0.730	10.5
12	Oakland Spiders	1	29	10.9%	13	16	44.8%	0.348	6.0

4 Discussion

4.1 What the patterns add to the leaderboard

The main contribution of these displays is interpretive rather than merely decorative. A possession path connects a metric to the sequence of throws that produced it. Eberhard's article *Six Players, Six Different Passer Types* makes that connection at the player level by showing that high aEC can arise from different combinations of deep throws, throwing volume, field-position decisions, and continuation options such as swings and resets (Eberhard, 2026c). Here, the same logic operates one level up. Empire's selected rows suggest a direct aggressive branch alongside longer diverse routes. Oakland's rows show why a team's most frequent family may not have the strongest OOE. Austin and Minnesota both carry short, huck-heavy routes beside longer possessions that contribute more of the chill flow up the field. Mean total aEC and median throws remain useful context, but neither is a universal quality score; each describes a different aspect of the route.

4.2 Turnovers change the pattern story

If only scoring possessions were retained, several of the selected overlays would look cleaner and several OOE values would become meaningless. Including turnovers changes the interpretation in two ways. First, it makes "how often this possession grouping ends in a goal" an observable outcome ratio rather than a description of successful examples only. Second, it lets spatially similar routes reveal where the same pattern breaks down. Therefore, the negative evidence is needed to distinguish a reliable pattern from a memorable one.

That is also why the manuscript reports n beside every percentage. A 100 percent group with two possessions is a promising observation. It is not a league-ready estimate. The public Shown Space writing repeatedly combines volume, risk, and role when telling player stories (Eberhard, 2026b,a); team patterns deserve the same discipline.

4.3 From recurring paths to descriptive groupings

Early in the project, the analysis tried to give each set of similar possessions a short tactical label. As more cards were reviewed, that approach became harder to apply consistently: two paths could share a broad route while differing in the exact throws, release points, or finishing sequence. The final groups therefore function first as spatial groupings. Their labels describe what the overlays share, while leaving more specific tactical interpretation for later review.

The productive next step is a second manual pass. Once a team has a small set of spatial families, an analyst can name them using both their geometry and their function: for example, a direct deep branch, a lateral continuation sequence, or a long reset-and-advance pattern. The figures make those judgments auditable. Another analyst can see the same overlays, disagree with a boundary, and revise the group without changing the underlying possession data.

5 Limitations

Several limitations should shape how these results are used.

First, the pattern taxonomy is analyst-curated. The three rows are useful descriptive units, but their boundaries reflect visual judgment and the current arrangement checkpoints. A future study could ask another analyst to classify the same cards and measure how often the two classifications agree. It could also compare these human-made rows with groups created from simple path coordinates and distances.

Second, the paper studies four teams and three selected rows per team. Large unsorted groups remain outside the narrative by design. The figures are not a complete description of any offense, and they should not be used to infer that an unshown pattern is rare without checking the full browser.

Third, the sample is regular season only. It does not answer whether a team changed its offense in the playoffs. Nor does the current summary adjust for opponent pressure, field conditions, score state, personnel, or starting field position. These factors can influence both the path and the outcome.

Finally, the paper uses the throw-level aEC values supplied in the local Shown Space play-by-play records; it does not refit or audit the underlying machine-learning model. The metric is treated as a contextual contribution signal, not as a causal estimate that a pattern alone produced the final score.

6 Conclusion

The four teams in this study do not reduce to a single offensive template. Empire's selected groups combine high OOE with a direct aggressive branch and longer middle-to-sideline routes. Austin carries a wide, patient attack beside shorter diagonal huck options. Minnesota keeps its selected paths compact and middle-heavy, emphasizing a clear team identity. Oakland makes the strongest case for heterogeneity: its largest selected family is turnover-heavy, while two smaller families are much more successful and flow-oriented.

Taken together, the findings answer the title's question: both deep hucks and small-ball sequences can work. The better choice depends on a team's personnel, the strengths those players combine, and the scoring opportunities that combination creates. The heatmap and overlays show how those choices affect possession length, field movement, and outcomes, while OOE and total aEC provide context without pretending that one number explains a route. The data do not identify a universal winning recipe; they show that teams are best served when their offensive design turns their particular connections into repeatable scoring opportunities.

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